

# HAW Hamburg

Informatik 1 Protokoll – WS2025/2026

Team 15

Labor 4

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## Aufgabe 1)

Unsere Überlegungen finden Sie im Programm.

Testfälle:

- Vater(degenhard\_duck, della\_duck) = true

```
 vater(degenhard_duck, della_duck).  
  
true  
  
Next 10 100 1,000 Stop  
  
?- vater(degenhard_duck, della_duck).
```

- Mutter(daphne\_duck, gustav\_gans) = true

```
 mutter(daphne_duck, gustav_gans).  
  
true  
  
?- mutter(daphne_duck, gustav_gans).
```

- oma(dankrade\_duck, donald\_duck) = true

```
 oma(dankrade_duck, donald_duck).  
  
true  
  
Next 10 100 1,000 Stop  
  
?- oma(dankrade_duck, donald_duck).
```

- Opa (emelrich\_erpel, X) X = teddy\_duck, daphne\_duck, degenhard\_duck, wilhelmine\_erpel, willibald\_wasserhuhn

```
X = teddy_duck  
X = daphne_duck  
X = degenhard_duck  
X = wilhelmine_erpel  
X = willibald_wasserhuhn  
  
?- opa(emelrich_erpel, X).
```

- Vorfahre(X, dorette\_duck) X = gunhilda\_gans, emelrich\_erpel, unbekannte\_frau\_von\_emil-erasmus, emil-erasmus\_erpel

**X** = gunhilda\_gans

**X** = emelrich\_erpel

**X** = unbekannte\_frau\_von\_emil-erasmus

**X** = emil-erasmus\_erpel

---

```
?- vorfahre(X, dorette_duck).
```

- Vollbruder (della\_duck, donald\_duck) = false

 vollbruder(della\_duck, donald\_duck).

false

---

```
?- vollbruder(della_duck, donald_duck).
```

- Vollschwester (della\_duck, donald\_duck) = true

 vollschwester(della\_duck, donald\_duck).

true

Next

10

100

1,000

Stop

---

```
?- vollschwester(della_duck, donald_duck).
```

- Onkel (jakob\_duck, X) X = dortel\_duck, dagobert\_duck, mathilda\_duck

 onkel(jakob\_duck, X).

**X** = dortel\_duck

**X** = dagobert\_duck

**X** = mathilda\_duck

---

```
?- onkel(jakob_duck, X).|
```

- Tante (dorette\_duck, X) X = wilhelmine\_erpel, willibald\_wasserhuhn

```
⚙️ tante(dorette_duck, X).  
  
X = wilhelmine_erpel  
X = willibald_wasserhuhn  
  
?- tante(dorette_duck, X).
```

- Cousin (willibald\_wasserhuhn, teddy\_duck) = true

```
⚙️ cousin(willibald_wasserhuhn, teddy_duck).  
  
true  
  
Next 10 100 1,000 Stop  
  
?- cousin(willibald_wasserhuhn, teddy_duck).
```

- Cousine (daphne\_duck, X) X = wilhelmine\_erpel, willibald\_wasserhuhn

```
⚙️ cousine(daphne_duck, X).  
  
X = wilhelmine_erpel  
X = willibald_wasserhuhn  
  
false  
  
?- cousine(daphne_duck, X).
```

## Aufgabe 2)



(Es bietet sich an, die Variante zu wählen, in welcher die 22 zur Wurzel wird, um den Baum balanciert zu halten.)

|    |   |    |   |    |    |    |   |   |    |    |    |
|----|---|----|---|----|----|----|---|---|----|----|----|
| 1  | 2 | 3  | 4 | 5  | 6  | 7  | 8 | 9 | 10 | 11 | 12 |
| 25 | 9 | 41 | 7 | 17 | 36 | 52 |   |   | 13 | 20 | 28 |

|    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|    |    | 60 |    |    |    |    |    |    |    | 22 |    |

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| 25 | 26 | 27 | 28 | 29 | 30 | 31 |
|----|----|----|----|----|----|----|

|  |  |  |  |  |  |  |
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|--|--|--|--|--|--|--|